Missing values in proteomics data and the limpa R package

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References

- Mengbo Li, Gordon K Smyth, Neither random nor censored: estimating intensity-dependent probabilities for missing values in label-free proteomics, Bioinformatics, Volume 39, Issue 5, May 2023, btad200, https://doi.org/10.1093/bioinformatics/btad200
- Li, M., Cobbold, S. A., & Smyth, G. K. (2025). Quantification and differential analysis of mass spectrometry proteomics data with probabilistic recovery of information from missing values. bioRxiv, 2025-04. https://doi.org/10.1101/2025.04.28.651125

Missingness in proteomics

- Shotgun proteomics uses mass spectrometry to measure all proteins present in a sample
- Proteins are fragmented into peptides that are identified by comparing their spectra to a library of theoretical spectra
- Missing values (where a peptide or protein is present in some samples and undetected in others) are common

Missingness in proteomics

- Missingness happens when
  1. The amount of a peptide/protein in the sample is below the (unknown) limit of detection (intensity dependent)
  2. A peptide's signature cannot be distinguished from that of other peptides in the sample (intensity dependent + random)
- Missing values are more common for less abundant peptides/proteins but can still occur at higher intensities

Example proteomics data

	Sample 1	Sample 2	Sample 3	Sample 4	Sample 5
proteoglycan 4	33428	31346	25916	31561	47280
von Willebrand factor	NaN	NaN	NaN	NaN	89
High affinity immunoglobulin epsilon receptor subu	119	84	107	98	98
SH3 domain-binding glutamic acid-rich-like protein	1220	3368	2685	2989	1411
Joining chain of multimeric IgA and IgM	2399	1584	5239	2285	2613
Adiponectin B;Complement C1q C chain	10071	20210	11344	11843	29717
Equ c 2 allergen;Lipocalin/cytosolic fatty-acid bi	27	NaN	NaN	NaN	NaN
Protein O-linked-mannose beta-1,2-N-acetylglucosam	227	253	284	256	354

Imputation

- Missing values in proteomics are often addressed by single imputation
  - Replace missing values with 0, lowest observed value, mean, other simple quantity
  - Imputation packages such as Perseus, MSImpute, MsCoreUtils and others
- Uncertainty in imputated observations not accounted for
- Data are MNAR

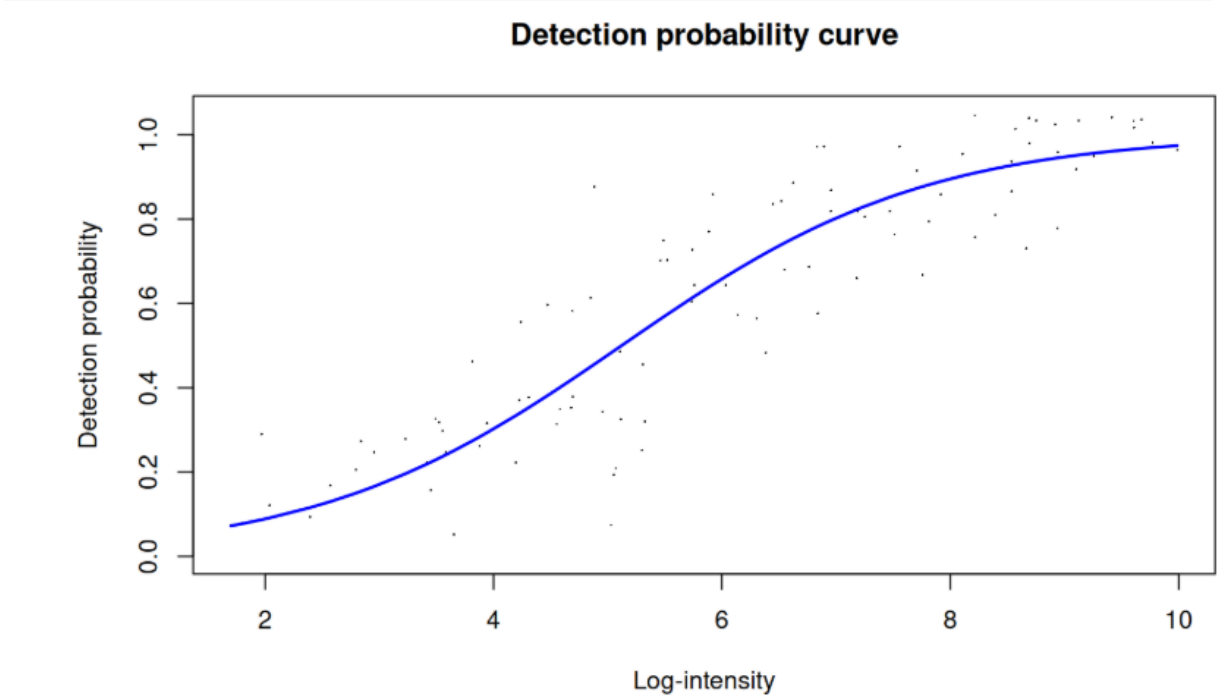
Left censoring

- Left censoring approach treats missing observations as censored at (unknown) detection limit
- Inappropriate if missingness isn’t strictly intensity-dependent

Filtering

- Proteins with above a certain number of missing values (e.g. 50% of samples) are often removed prior to analysis
- This helps but does not eliminate all missing values

The detection probability curve



The detection probability curve

- Li and Smyth (2023 & 2025) use a logit linear model for the probability of missingness, given the (possibly unobserved) intensity
- log intensity (observed and unobserved values) assumed to be normally distributed (“complete normal”)

The DPC model

- y is log intensity for a given peptide in a given sample
- d = 1 if y is observed, 0 otherwise

$$y \sim N(\mu, \sigma^2)$$

$$f(d = 1|y) = \text{logit}^{-1}(\beta_0 + \beta_1 y)$$

The DPC model

- The marginal probability for the missing values is obtained by integrating over y

$$f(d=0) = \int f(d=0|y)\phi(y;\mu,\sigma^2)dy$$

$$= \int [1 - f(d=1|y)]\phi(y;\mu,\sigma^2)dy$$

$$= \int [1 - \text{logit}^{-1}(\beta_0 + \beta_1 y)]\phi(y;\mu,\sigma^2)dy$$

- where ϕ is the normal pdf

The DPC model

- The log likelihood (for data from a given protein) is then

$$\sum_{j=1}^{P}\sum_{i=1}^{n} I\left(d_{ij}=1\right)\log[\text{logit}^{-1}(\beta_0+\beta_1 y)]$$
$$+ I\left(d_{ij}=0\right)f\left(d_{ij}=0\right)$$

where $j = 1, \dots, P$ are the peptides for a given protein and $i = 1, \dots, n$ are the samples

- Parameters of the DPC are thus obtained by MLE using Gaussian quadrature.

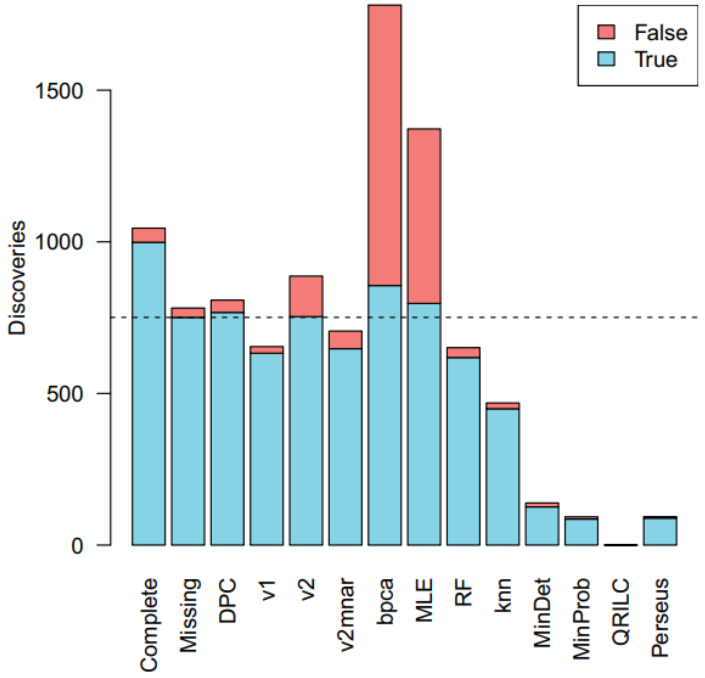
Quantifying protein expression

- The DPC is fitted to data at the peptide precursor level
- For differential expression analysis, we need data at the protein level

Quantifying protein expression

- Protein expression is summarized from the peptide data and the DPC using empirical Bayes
- Each observation has an associated variance weight for downstream modelling
  - Proteins with more missing data are downweighted
  - Proteins with fewer peptide precursors are downweighted
- Input to the DE model is protein-level quantification + variance weights

Results on simulated data using limpa R package



Results on simulated data using limpa R package

- Imputation methods (v1 through Perseus in figure) either inflate the FDR or have reduced power
- Limpa preserves FDR, slightly better power than just keeping missing values missing
- Impact of method likely greater for higher proportion of missing values
- Can still be used when an entire experimental group is missing data for a given protein

Real data example

- Plasma samples from 25 horses with and without osteoarthritis
- Preprocessing and peptide identification conducted in Spectronaut software
- 60K peptides identified

Real data example

```
library(limpa)
y.peptide <- readSpectronaut(file =
file.use, run.column = "R.FileName",
extra.columns = c("PG.ProteinAccessions",
"PG.ProteinDescriptions"), q.cutoffs =
rep(0.01, 2))
```

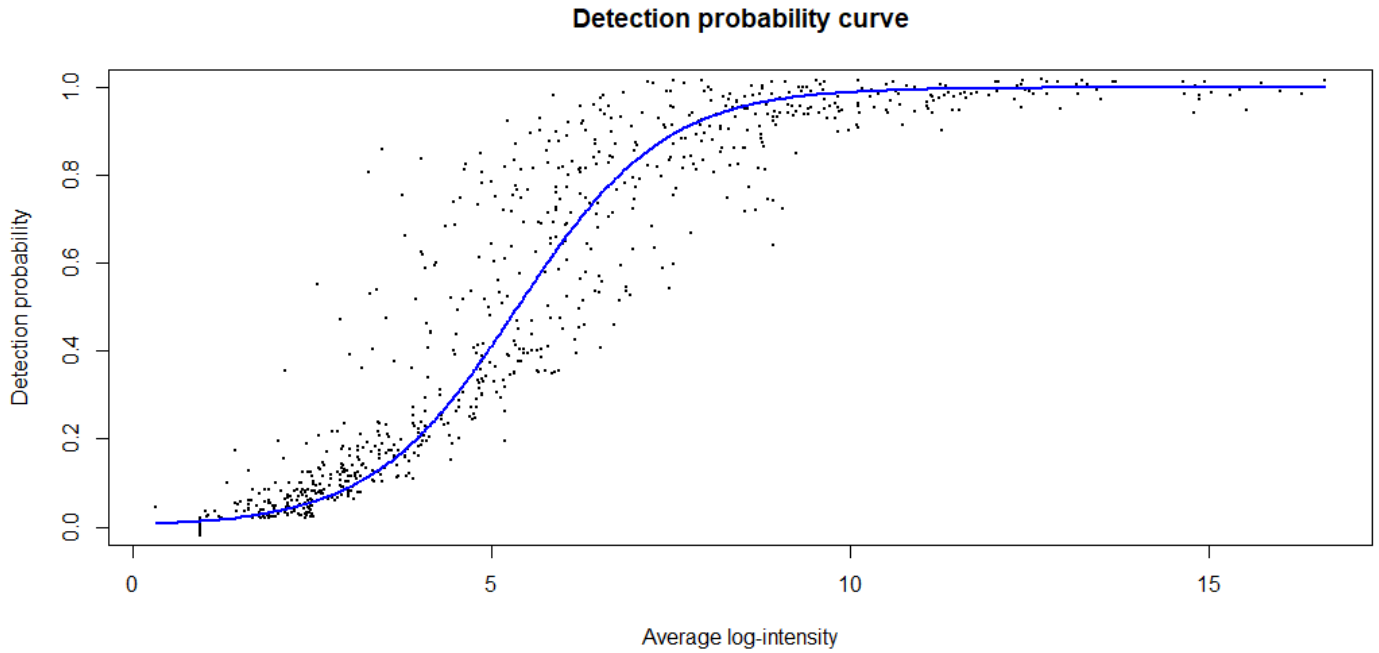
Real data example

```
y.peptide <-
filterCompoundProteins(y.peptide,
protein.group = "PG.ProteinAccessions")
y.peptide <-
filterSingletonPeptides(y.peptide,
min.n.peptides = 2, protein.group =
"PG.ProteinAccessions")
```

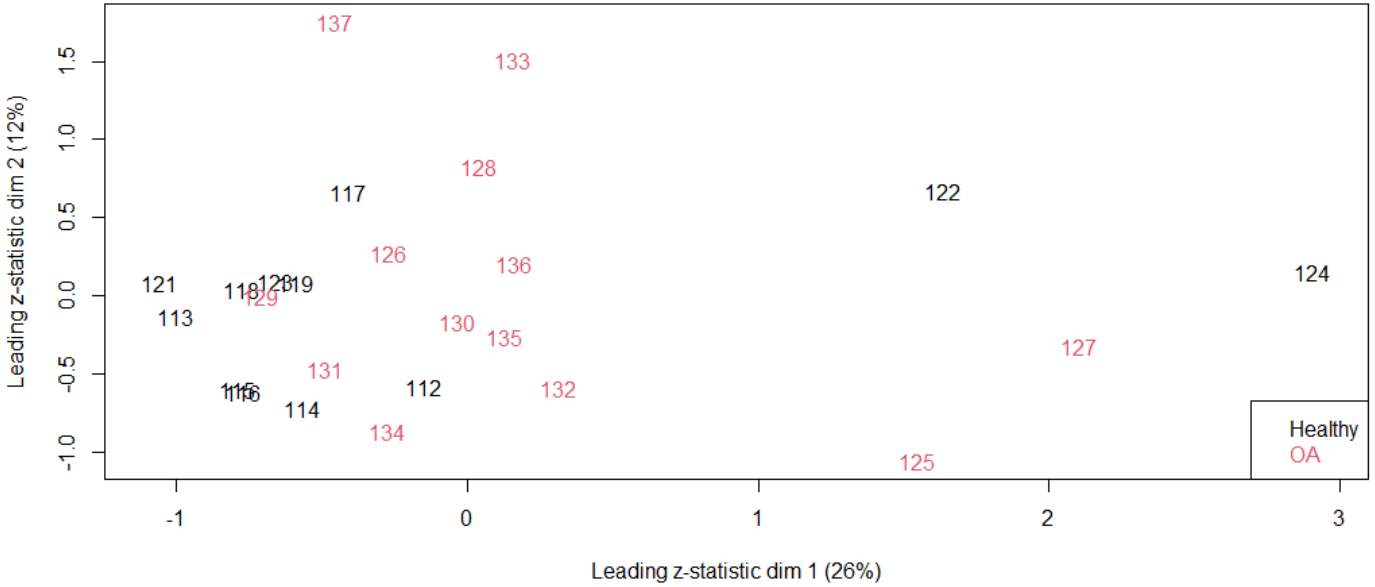
Real data example

```
dpcfit <- dpcCN(y.peptide)
y.protein <- dpcQuant(y.peptide,
"PG.ProteinAccessions", dpc = dpcfit)
```

Real data example



Real data example



Real data example

```
design <- model.matrix(~`Healthy vs OA` + `Age (years)` + Sex + `Breed 2`, data = pdata)
fit <- dpcDE(y.protein, design, plot = FALSE)
fit <- eBayes(fit)
```

Real data example

PG.ProteinAccessions	PG.ProteinDescriptions	NPeptides	PropObs	logFC	P.Value	adj.P.Val
F7DTP2	Glutathione reductase	21	0.14	-2.38	1.6e-05	0.023
A0A3Q2HM37	Periostin	3	0.25	-1.54	2.9e-05	0.023
F7DMX6	Insulin-like growth factor-binding protein 1	11	0.47	2.75	6.5e-05	0.027
F6SX07	Galectin	15	0.07	-2.39	7.3e-05	0.027
F6VJF5	SH3 domain-containing protein	13	0.12	-2.32	8.4e-05	0.027
A0A9L0RK94	Uncharacterized protein	3	0.65	-1.17	1.0e-04	0.027
A0A9L0RI10	Jupiter microtubule associated homolog 1	6	0.12	-1.63	0.00012	0.028
F7BQT4	5,6-dihydroxyindole-2-carboxylic acid oxidase	8	0.54	-2.23	0.00023	0.047
F7CN11	Fibronectin	3	0.55	-1.52	0.00028	0.050